

About

INSTROLE offers real-time monitoring and record of precise, quantitative measures of one's bipedal weight distribution during quiet standing and dynamic gait for young and elderly with and without balance disorders.

People

Prof. Uttama Lahiri
Director (Technology)

Dr. Neelesh Kr. Sharma
Director (Design)

Contact



+91- 92135 99040



oreejoy@gmail.com



<https://oreejoy.in/>



Room No. 302, IIT
Gandhinagar Research Park



IIT Gandhinagar



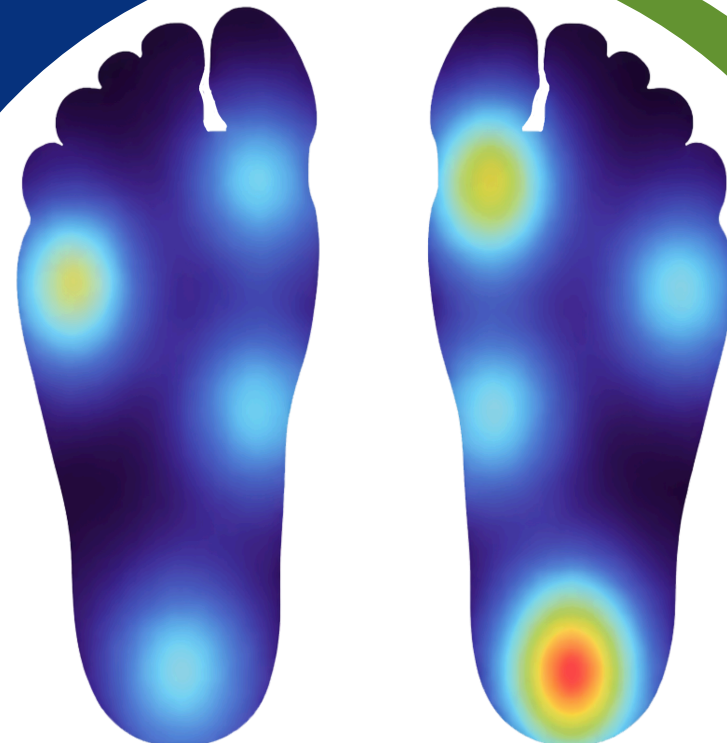
Sarita G Gelot Laboratory for
Intelligent Rehabilitation &
Affective Computing Systems

The Next Generation INSTRUMENTED SMART SHOE INSOLE



INSTROLE

Partner for a healthy balance and gait



How to use INSTROLE ?



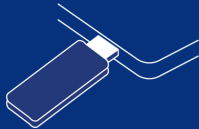
Insert the smart instrumented sole INSTROLE into a shoe



Follow GUI provided with INSTROLE on tablet PC. Collect and store output: gait indices, weight distribution for quiet standing and dynamic gait conditions



Establish and verify connections from INSTROLE to tablet PC



Get a consolidated report for scrutiny by medical expert



Features

- Sole with Embedded sensors connected to a microcontroller for data acquisition.
- Wireless connectivity of microcontroller with tablet PC.
- Measures and records weight distribution while at rest and in motion assisting in diagnostics such as flat foot.
- Measures and quantifies gait in terms of step time, stride time, cadence, stance and swing phases, etc.
- Suitable for monitoring over any ground surface or treadmill-assisted walk/run.

Benefits

Complement clinical diagnosis by offering objective metrics while saving administration time

Offer visualization of balance and gait conditions to patients to realize the possible risk factors

Monitor patient's progress thereby motivating the patient to undergo treatment

Quantified screening for early diagnosis to prevent possible future risks

Easy quantification of weight distribution with consolidated report

User-friendly system of hardware and software

Specifications



Technical

08
Sensors per pair

50 Hz
Sampling rate

99.2 %
Accuracy

150 g
Weight


Connectivity


Data export

06 hr
Battery life

Orthopaedics



Pre and post TKA
symmetry analysis



UKR load
distribution

Neurology



Stroke gait
asymmetry



Parkinson's step
variability



Fall risk
screening

Geriatrics

Reporting

Plantar pressure heat map



Step time variability chart



Symmetry index (%)



 Room No. 302, IIT Gandhinagar Research Park



Prof. Uttama Lahiri

Director (Technology)
Alumnus, IIT Kaharagpur

 uttamalahiri@iitgn.ac.in

 +91- 7923952435



Dr. Neelesh Kr. Sharma

Director (Design)
Alumnus, IIT Patna

 neelesh.sharma@iitgn.ac.in

 +91- 7992063640

